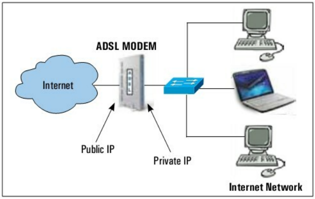

Diagram 1: Basic network

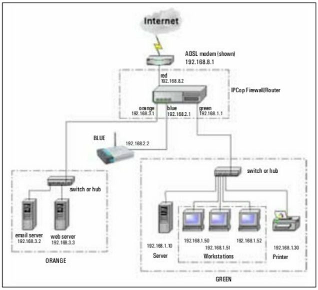

Diagram 2: A typical firewall installation

Operation

A typical network firewall is installed between the internal network and the Internet; thus, all traffic between them passes through the firewall. For Internet connections via an ADSL modem, the firewall will be installed between the internal network and the ADSL modem, whereas for an Internet connection with an Ethernet connection, the firewall will replace the modem or router as depicted in Diagram 2. The firewall analyses everything passing through, and based on the configured policy, let's through only 'safe' traffic.

Various client computer systems request data from application servers simultaneously. A 'port' number is used to differentiate this traffic. A client requesting data from the server uses the destination port number of the corresponding service. For example, a computer system requesting HTTP data will use port 80.

The traffic could use one of the following two protocols:

- Transmission Control Protocol (TCP), which guarantees delivery of the data, is reliable but has larger headers to accommodate the handshake signals and flags required for assured delivery.
- User Datagram Protocol (UDP), which does not guarantee delivery of the data, but has a smaller header. Here, higher-level protocols may take care of assured delivery.

Both TCP and UDP protocols have 65535 ports each (2 to the power of 16, port 0 unused). Ports up to 1024 are reserved. For example, well known TCP ports are HTTP (80), HTTPS (443), FTP (21), telnet (23), whereas UDP port 53 is reserved for the DNS service.

Caution: https://IPCOP:8443/cgi-bin/index.cgi will open URL Filter screen in IPCop 2.0.6. However, please be careful while configuring, some of the settings will not function as expected. Fully functional URL Filter is expected in IPCop 2.1 version.

A firewall uses this port number to identify the traffic. In the HTTP example mentioned above, the firewall reads HTTP traffic to port 80, and passes it to the Internet only if it matches the desired policy. Unwanted websites and content as defined in the policy will be dropped. Typical functions of a network firewall can be classified into traffic control and others, as shown below.

Traffic control functions are:

- Access from the Internet to the internal network
- Website access from the internal network to the Internet
- Download of various file types such as audio/video
- Port-wise access from the internal network to the Internet
- Bandwidth control

Since all traffic between the internal network and the Internet passes via the firewall, it is the best point to provide various other functions such as:

- A VPN gateway between two networks connected via the Internet
- A VPN server for remote clients connecting to the internal network
- Authentication of local users for Internet access
- Generation of traffic graphs
- Logging Internet access

Why IPCop?

For a long time, the open source community has provided many options for network firewalls by releasing various distributions. They provide security and ease of configuration, and can be installed on practically any minimal-configuration computer system. The most important factor for SMBs is that these distros are free (under a GNU license) and do not require yearly renewals. One of the best of these is the IPCop firewall, which has a long history—it was forked from Smoothwall in 2001. Various releases followed, the most popular being IPCop version 1.4.21 (the last stable version available).

The default v 1.4.21 had limited functionality, but was flexible enough to allow installation of various add-